CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 14: Rectilinear Figures

Topic-wise short notes with solved examples

Polygons and their angle sums, and the properties of quadrilaterals: parallelogram, rectangle, rhombus, square, trapezium and kite, with the conditions that identify each.

1 Polygons

- A **rectilinear figure** is a closed plane figure bounded by line segments. A **polygon** has three or more sides.
- A polygon is **convex** if every interior angle is less than 180°. It is **regular** if all its sides are equal and all its angles are equal.
- For a convex polygon of n sides:

Quantity	Formula
Sum of the interior angles	$(2n-4)$ right angles $= (n-2)\times 180^\circ$
Sum of the exterior angles	360°
Each interior angle of a regular polygon	$\dfrac{(2n-4)\times 90^\circ}{n}$
Each exterior angle of a regular polygon	$\dfrac{360^\circ}{n}$
Number of diagonals	$\dfrac{n(n-3)}{2}$

- At each vertex, interior angle $+$ exterior angle $= 180^\circ$.

Example 1.1 Find the sum of the interior angles of a hexagon and each angle of a regular hexagon.

SOLUTION

$(6-2)\times 180^\circ = 720^\circ$; each angle $= 720^\circ \div 6$

720°; 120°

Example 1.2 Each interior angle of a regular polygon is 150°. Find the number of sides.

SOLUTION

Each exterior angle is 30°, so $n = 360 \div 30$.

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Example 1.3 The angles of a pentagon are in the ratio $4 : 8 : 6 : 4 : 5$. Find them.

SOLUTION

The sum is 540° and there are 27 parts, so one part is 20°.

$80^\circ, 160^\circ, 120^\circ, 80^\circ, 100^\circ$

Example 1.4 The ratio of an interior angle to an exterior angle of a regular polygon is $7 : 2$. Find the number of sides.

SOLUTION

$7k + 2k = 180^\circ$, so the exterior angle is 40°.

9 sides

2 Quadrilaterals

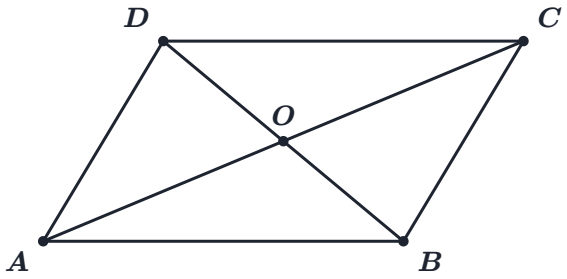
- The sum of the angles of a quadrilateral is 360°.

Quadrilateral	Definition
Trapezium	One pair of opposite sides parallel
Isosceles trapezium	A trapezium with equal non-parallel sides
Parallelogram	Both pairs of opposite sides parallel
Rectangle	A parallelogram with one angle 90°
Rhombus	A parallelogram with a pair of adjacent sides equal
Square	A rectangle with adjacent sides equal
Kite	Two pairs of equal adjacent sides

3 Parallelogram

- Opposite sides are equal.
- Opposite angles are equal.
- Adjacent angles are supplementary.
- The diagonals bisect each other.
- Each diagonal divides it into two congruent triangles.

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Parallelogram ABCD: AO = OC and BO = OD

- **A quadrilateral is a parallelogram if** any one of these holds: both pairs of opposite sides are equal; both pairs of opposite angles are equal; the diagonals bisect each other; **one pair of opposite sides is equal and parallel**.

Example 3.1 In parallelogram ABCD, $\angle A = (3x + 10)^\circ$ and $\angle B = (2x - 5)^\circ$. Find all the angles.

SOLUTION

$3x + 10 + 2x - 5 = 180$, so $x = 35$.

$\angle A = \angle C = 115^\circ$ **and** $\angle B = \angle D = 65^\circ$

Example 3.2 In parallelogram ABCD, the diagonals meet at O. If $AO = 2x + 1$, $OC = 3x - 4$, find AC.

SOLUTION

$2x + 1 = 3x - 4$, so $x = 5$ and $AO = 11$

$AC = 22$

4 Rectangle, rhombus and square

Property	Rectangle	Rhombus	Square
All angles 90°	Yes	No	Yes
All sides equal	No	Yes	Yes
Diagonals equal	Yes	No	Yes
Diagonals perpendicular	No	Yes	Yes
Diagonals bisect the angles	No	Yes	Yes

- All three have every property of a parallelogram.
- In a rectangle the diagonals are equal and bisect each other, so $OA = OB = OC = OD$.
- In a rhombus each diagonal is the perpendicular bisector of the other.
- In a square each diagonal makes 45° with the sides.

Example 4.1 In rectangle ABCD the diagonals meet at O and $\angle AOB = 124°$. Find $\angle OAB$ and $\angle OBC$.

SOLUTION

$OA = OB$, so $\angle OAB = \dfrac{180° - 124°}{2} = 28°$.

$\angle OBC = 90° - 28°$

$28°$ **and** $62°$

Example 4.2 In rhombus ABCD, $\angle A = 60°$ and the side is 10 cm. Find the diagonals.

SOLUTION

Triangle ABD is equilateral, so $BD = 10$ cm. Half of AC is the altitude: $\sqrt{100 - 25} = 5\sqrt{3}$.

$BD = 10$ **cm and** $AC = 10\sqrt{3}$ **cm**

Example 4.3 ABCD is a square. E is a point outside the square such that ABE is an equilateral triangle. Find $\angle DEC$.

SOLUTION

$\angle DAE = 90° + 60° = 150°$ and $AD = AE$, so $\angle AED = 15°$. Similarly $\angle BEC = 15°$.

$\angle DEC = 60° - 15° - 15°$

$30°$

5 Trapezium and kite

- **Trapezium** with $AB \parallel DC$: $\angle A + \angle D = 180°$ and $\angle B + \angle C = 180°$.
- **Isosceles trapezium**: base angles are equal ($\angle A = \angle B$, $\angle C = \angle D$) and the diagonals are equal.
- **Kite** with $AB = AD$ and $CB = CD$: diagonal AC is the perpendicular bisector of BD and bisects $\angle A$ and $\angle C$; $\angle B = \angle D$.

Example 5.1 In trapezium ABCD, $AB \parallel DC$, $\angle A = 72°$ and $\angle C = 115°$. Find $\angle B$ and $\angle D$.

SOLUTION

$\angle D = 180° - 72°$ and $\angle B = 180° - 115°$

$\angle B = 65°$, $\angle D = 108°$

Example 5.2 In isosceles trapezium ABCD with $AB \parallel DC$ and $AD = BC$, $\angle A = 70°$. Find the other angles.

SOLUTION

$\angle B = 70°$, $\angle C = \angle D = 110°$

6 Proofs with parallelograms

Example 6.1 ABCD is a parallelogram. E and F are the mid-points of AB and CD. Prove that AECF is a parallelogram.

SOLUTION

$AB \parallel DC$ and $AB = DC$. So $AE = \frac{1}{2}AB = \frac{1}{2}DC = FC$, and $AE \parallel FC$.

One pair of opposite sides is equal and parallel, so AECF is a parallelogram

Example 6.2 Prove that the bisectors of two adjacent angles of a parallelogram meet at right angles.

SOLUTION

$\angle A + \angle B = 180^\circ$, so $\frac{1}{2}\angle A + \frac{1}{2}\angle B = 90^\circ$.

In the triangle formed with side AB, the third angle is $180^\circ - 90^\circ = 90^\circ$

7 Common mistakes

- Using $(n-2) \times 180^\circ$ for the sum of the **exterior** angles. That sum is always 360°.
- Saying the diagonals of a parallelogram are equal or perpendicular. That is true only for special types.
- Treating adjacent angles of a parallelogram as equal.
- Assuming a quadrilateral with one pair of equal sides and one pair of parallel sides (different pairs) is a parallelogram.
- Forgetting that a rhombus and a rectangle are both parallelograms.

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